

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1711

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

*I **B.Bhargavi 192311136**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : B.Bhargavi

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

Full Citation:

Neves, M., Vieira, M., & Neto, P. (2021). *A study on a Q-Learning algorithm application to a manufacturing assembly problem*. **Journal of Manufacturing Systems**, **59**, 426–440. <https://doi.org/10.1016/j.jmsy.2021.02.014>.

The application domain is **manufacturing and industrial assembly**. The research focuses on the problem of deciding the best order in which assembly tasks should be performed. In manufacturing, different task sequences may be feasible, but they can require different amounts of time. Therefore, selecting an efficient sequence can reduce production time.

The authors formulate the assembly problem as a **Markov Decision Process (MDP)** and apply **model-free Q-Learning**. The Q-Learning agent learns the value of different state-action combinations through repeated interaction with the assembly environment.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

The main objective of the article is to investigate whether Q-Learning can automatically learn an efficient assembly sequence for a manufacturing product while reducing the total assembly time. Traditional assembly optimization can become difficult because the number of possible task sequences grows rapidly as the number of tasks increases. The authors identified a gap in applying reinforcement learning to assembly-sequence optimization while considering task precedence, processing-time variations, reward design, and the exploration-exploitation problem. Their work addresses this gap by representing assembly as a Markov Decision Process and using a model-free Q-Learning algorithm to learn an optimal state-action policy. An aeroplane toy containing eight assembly tasks was used as the case study. The researchers tested increasingly complex scenarios, including estimated task times, measured task times, tool-changeover effects, and restricted impossible actions. They also performed sensitivity analyses on learning rate, discount factor, epsilon decay, reward parameters, and the number of episodes. The final restricted-action scenario achieved an optimal assembly sequence in 98.3% of the experiments while requiring only 780 episodes, demonstrating the potential of Q-Learning for manufacturing optimization.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

The main technique used in the article is **Q-Learning**, which is a **model-free, value-based reinforcement learning algorithm**.

The agent interacts with an environment according to the following process:

State → Action → Reward → New State

For every state-action pair, a Q-value is maintained. The Q-value represents the expected future reward obtained by selecting an action from a particular state.

The authors use the standard Q-Learning update:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- α = learning rate
- γ = discount factor
- r = reward received
- s = current state
- a = selected action
- s' = next state

The action-selection strategy is **epsilon-greedy**. With probability ϵ , the agent explores a random action; otherwise, it selects the action having the highest Q-value. Epsilon is gradually reduced using an epsilon-decay parameter.

Dataset / Case Study

This paper does not use a conventional tabular machine-learning dataset such as Iris or MNIST. Instead, it uses a **manufacturing assembly case study** based on an **aeroplane toy from the Yale-CMU-Berkeley (YCB) Object and Model Set**.

The assembly process was divided into:

- **8 assembly tasks/actions**
- **9 structural parts**
- **2 types of fasteners**
- **40320 possible task sequences** theoretically, because $8! = 40320$
- **3360 feasible sequences** after considering task-precedence constraints.

The state represents the current assembly status. Initially, no task is completed, represented by 00000000. The final state is 11111111, representing completion of all eight tasks.

The authors initially used estimated task-processing times. In a more realistic scenario, task times were measured **10 times** and average values were calculated. The measured mean task times were approximately:

Task	Mean Time (time units)
1	5.93
2	8.16
3	8.99
4	7.48
5	9.88
6	10.38
7	11.48
8	8.82

The second scenario also considered tool-changeover time, increasing the complexity of the state representation.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

The paper is directly related to the **Reinforcement Learning** portion of CO 4.

CO 4 Concept	Connection with Article
Reinforcement Learning	Agent learns through interaction with the environment
MDP	Assembly problem is represented using states, actions, transitions and rewards

Q-Learning		Q-table stores the expected value of state-action combinations
Exploration vs Exploitation		Epsilon-greedy strategy balances random exploration and best-known actions
Reward Function		Rewards represent the quality/time efficiency of an assembly sequence
Policy		Final policy selects the assembly action with the highest Q-value

Methodological Strength

A major strength is the **systematic sensitivity analysis**. The authors did not simply run Q-Learning once. They studied learning rate, discount factor, epsilon decay, reward shift, reward multiplier, maximum steps, and maximum episodes. They also performed **120 experiments** for parameter settings to obtain statistically meaningful results.

Methodological Limitation

The major limitation is that the evaluation is based on **one assembly product and a controlled computational environment**. Although the authors use measured task times, the final result is not demonstrated on a physical industrial assembly line. Therefore, the reported performance may not directly represent performance under real-world disturbances such as sensor noise, robot failures, worker variation, unexpected delays, or changing product designs.

The authors themselves acknowledge that increasing assembly complexity can make standard Q-Learning impractical because of the **curse of dimensionality**.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

The researchers first created a mathematical optimization model as a baseline. The mathematical model found the optimal assembly sequence:

1 → 8 → 4 → 7 → 5 → 2 → 6 → 3

with an assembly time of 65 time units. The optimization model achieved an optimality gap of 0 in less than one second.

The Q-Learning experiments were then conducted under increasingly realistic conditions.

Comparative Results

Experiment	Main Condition	Maximum Episodes	Result
Mathematical optimization	MILP baseline	—	Optimal time = 65 t.u.
Q-Learning Scenario I	Estimated task times	Up to 6500	Learned optimal sequence $\approx 95.8\%$ of 120 experiments
Q-Learning Scenario II	Measured task times + tool changeover	6500	115/120 optimal $\approx 95.8\%$
Q-Learning Scenario III	Measured times + restricted impossible actions	780	98.3% optimal rewards

In Scenario II, using the selected parameter set and 6500 episodes, the agent learned one of the two optimal assembly sequences in 115 out of 120 experiments (approximately 95.8%). It learned a second-best sequence four times and a fifth-best sequence once, with no failure to produce a feasible sequence.

The most important result came from Scenario III. By restricting impossible actions instead of allowing the agent to select and penalizing them, the required maximum number of episodes was reduced to 780, while the percentage of optimal rewards increased to 98.3%. The 780 episodes correspond to approximately 21.7% of the feasible assembly sequences.

Best Q-Learning Parameters

Parameter	Value
Learning rate	1
Discount factor	1
Initial epsilon	0.9
Epsilon decay	0.00005
Maximum steps/episode	15
Maximum episodes	6500 in Scenario II
Reward shift	9
Reward multiplier	13
Reward penalty	-1,000,000

For the final restricted-action experiment, the maximum episode count was reduced to 780 and the reward penalty was removed because impossible actions were no longer available.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

Are the reported metrics realistic?

The reported **98.3% optimal-sequence rate** is impressive, but it should not be interpreted as a general real-world "98.3% accuracy." It means that the learned policy selected an optimal assembly sequence in 98.3% of the tested experiments under the particular simulated/modelled assembly environment.

This distinction is important because the experiment has:

- One product.
- Eight tasks.
- Known precedence constraints.
- Controlled processing-time information.
- A predefined reward function.
- A limited state space.
- Computational rather than full physical deployment.

Therefore, the result is realistic **for the experimental setup**, but additional validation is required before claiming industrial reliability.

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

The proposed approach can potentially be used in:

- Automated assembly lines
- Robotic manufacturing
- Smart factories
- Production scheduling
- Flexible manufacturing systems
- Human-robot collaborative assembly
- Customized product manufacturing
- Industrial work-cell optimization

For example, an industrial robot could observe the current assembly state and use a learned policy to determine which assembly operation should be performed next.

Challenges in Industrial Scaling

Data availability:

Although RL requires less labelled data than supervised learning, it needs many interactions with the environment. Physical experiments can be expensive because they consume materials and machine time.

Computational complexity:

As the number of assembly tasks increases, the number of states and possible actions grows rapidly. The authors specifically identify the curse of dimensionality as a limitation of Q-Learning.

Safety:

An incorrectly learned action could damage a component, machine or robot. Therefore, industrial RL systems require safety constraints.

Changing environments:

Manufacturing conditions can change because of machine failures, new products, worker behaviour and material availability.

Ethical and operational concerns:

Factories must maintain human safety, explainability, accountability and reliable operation when AI makes production decisions.

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

4(a) Three Key Insights

Insight 1: RL learns from interaction rather than labelled examples

The article helped me understand that reinforcement learning is fundamentally different from supervised learning. Instead of providing a labelled dataset containing the correct assembly sequence, the agent interacts with the environment and learns from rewards.

CO 4 Connection: Reinforcement Learning and MDP.

Insight 2: Reward design is extremely important

I learned that an RL algorithm does not automatically know what "good behaviour" means. The reward function determines what behaviour the agent should prefer. In this paper, reward values are connected to assembly time, and the authors show that reward shifts, penalties and multipliers can significantly affect learning performance.

CO 4 Connection: Reward function and Q-Learning.

Insight 3: Exploration and exploitation must be balanced

The article improved my understanding of the exploration-exploitation dilemma. If the agent only explores, it wastes many episodes trying poor actions. If it only exploits early, it may select a suboptimal sequence and never discover a better one.

The epsilon-greedy strategy provides a practical way to balance these two behaviours. The paper also demonstrates that epsilon decay strongly influences the number of episodes required for learning.

CO 4 Connection: Epsilon-greedy policy, exploration and exploitation.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Proposed Experiment: Deep Q-Learning for Multi-Product Robotic Assembly

I would extend the research by replacing the standard Q-table with a Deep Q-Network (DQN) and testing it on multiple YCB objects rather than only one aeroplane toy.

The input would describe:

Current assembly state
Completed tasks
Available parts
Tool currently attached
Processing-time estimates
Task precedence constraints

The DQN would estimate Q-values using a neural network instead of storing every state-action value in a table.

This extension is feasible because the original research already demonstrates that increasing the number of states can become a limitation for conventional Q-Learning. In the more complex scenario, the model has 511 possible state representations, with 149 feasible states.

The expected impact would be greater scalability to larger and more complicated products. The system could potentially generalize its learned knowledge across different assembly configurations instead of learning an independent Q-table for every product.

Overall Review

This article provides a clear and practical application of **Reinforcement Learning and Q-Learning** to manufacturing assembly optimization. Its strongest contribution is showing that an RL agent can learn efficient assembly sequences without requiring a traditional labelled training dataset. The study is also valuable because the authors investigate important RL issues such as reward design, exploration-exploitation, epsilon decay and episode count.

The final result of **98.3% optimal assembly sequences** demonstrates strong performance within the experimental environment. However, the result should not be treated as direct evidence of 98.3% industrial accuracy because the study focuses on a single controlled case study. More products, physical robots, noisy environments and unseen assembly configurations should be tested.

Overall, the article significantly strengthens my understanding of **CO 4 – Reinforcement Learning**, particularly the concepts of MDPs, Q-values, reward functions, policies, exploration-exploitation and convergence. It also demonstrates how a theoretical machine-learning algorithm can be transformed into a practical optimization solution for smart manufacturing.

Reference

Neves, M., Vieira, M., & Neto, P. (2021). *A study on a Q-Learning algorithm application to a manufacturing assembly problem*. **Journal of Manufacturing Systems**, **59**, 426–440. DOI: 10.1016/j.jmsy.2021.02.014.

Publisher / DOI: [Elsevier ScienceDirect article](#)

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand